Hammad Rizwan

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Halifax, Canada

Education

Dalhousie University Sept. 2023 –
PhD Computer Science, Advisor: Hassan Sajjad

Lahore University of Management Sciences (LUMS) Sept. 2018 – Jun 2020
MS Computer Science, Advisor: Asim Karim

Research Interests

Model Editing, Machine Unlearning, Interpretability and Self-supervised Learning.

Skills

ML/AI: PyTorch, TensorFlow, HuggingFace, Transformers, Scikit-learn, LLM Fine-tuning, LoRA/Adapters, Self-Supervised Learning, Rasa
Programming: Python, C, Java, Go, Bash
Data & Analysis: NumPy, Pandas, Scipy, Matplotlib, Seaborn, Jupyter
Backend: FastAPI, Flask, REST API, Gunicorn, Postman
Databases / Retrieval: Qdrant, FAISS, MongoDB, Redis, Elasticsearch
MLOps / Deployment: Docker, HPC/Slurm, Conda/Virtual Environments
Systems & Tools: Linux, Git, LaTeX

Industry Experience

Upbeing.ai **Mitacs Accelerate Intern** Feb. 2024 – Nov. 2024

- Built and deployed a conversational AI system for a mental health support application, **combining large language model (LLM) based dialogue models, retrieval augmented generation (RAG),** and **representation similarity search pipelines** to deliver accurate, context-aware responses.
- **Collaborated with cross-functional team of researchers** to design and integrate model evaluation workflows, improving system reliability and advancing **AI-driven mental health research**

Arbisoft **Senior ML Engineer** Feb. 2023 – July 2023

- Built an end-to-end **personalized, machine learning–driven recommendation system** for a dining-focused social media platform, leveraging embedding models to **improve user engagement and content discovery**.
- **Optimized entity extraction** on noisy short-text inputs, increasing downstream **real-time search relevance** and **recommendation quality**.

ISSM.ai **NLP Engineer** Oct. 2022 – Feb. 2023

- Built **enterprise conversational AI systems** (chatbots) for UBL and Allied Bank, applying **intent recognition, dialogue management, and information retrieval** techniques to improve query resolution time and enhance customer experience.
- Developed and deployed **multilingual NLP pipelines** for AI-driven automation of banking workflows in Urdu/Roman Urdu, **training domain-specific embedding LLMs** and leveraging transliteration techniques to reduce manual processing and improve operational efficiency.

Eighty.ai **NLP Engineer** Jan. 2021 – Dec. 2021

- Designed and deployed a **machine learning–driven HS code classification** pipeline, tackling a **large-scale, fine-grained prediction** task spanning 97 categories and 1,244 subcategories, and improving **accuracy and scalability** for **global trade and compliance workflows**.
- Trained **NER models and domain-specific embedding LLMs** to build a high-accuracy document processing pipeline for the financial domain, enabling **automated information extraction, entity normalization, and downstream compliance automation**.

Ai XPRT (AuditXPRT) **NLP Engineer** July 2020 – Dec. 2020

- Developed a risk detection pipeline for **AIA Insurance**, **leveraging entity resolution and name-matching algorithms** to flag high-risk individuals and corporations for anti–money laundering (AML) compliance monitoring.
- Enhanced ComplyXPRT, a regulatory compliance platform for **Barclays Bank**, by integrating **continuous learning** for adaptive modelling and reducing inference latency by 30% through **model distillation, optimized deployment, and multi-threaded inference**.

Research Experience

HyperMatrix Lab Sept. 2023 –

Doctoral Researcher

- Investigating **knowledge editing in LLMs**, with a focus on lexical bias disentanglement and weight-preserving parameterization to enable controlled factual updates while minimizing collateral interference in unrelated knowledge.
- Designing **instance-level difficulty frameworks for machine unlearning**, leveraging sample-level hardness estimation.
- Contributing to **interpretability research**, including **range-based attribution methods** that characterize neurons over activation intervals, and ongoing work on **model pruning** for circuit discovery.

Robotics and Intelligent Computing (RICE) Lab Nov. 2020 – June 2022

Research Associate, Digital Forestry

- Designed a **hybrid vision pipeline** for monocular tree diameter estimation, combining **deep learning–based segmentation with classical feature extraction and regression-based post-processing**, addressing challenges of unstructured outdoor forestry imagery.
- Created the first **annotated dataset for tree classification and semantic segmentation** in Pakistan, supporting downstream research and deployments by **WWF and Forestry Pakistan**.
- Contributed to scalable, low-cost computer vision systems for national-scale digital forestry initiatives.

Knowledge and Data Engineering (KADE) Lab Sept. 2019 – June 2020

Research Assistant, Hate Speech and Generative Models

- Created the first **open-source datasets for offensive and hate speech detection in Roman Urdu**, and introduced a **custom CNN–ngram model** that established strong baselines; the work has since been **cited over 100 times** and serves as a benchmark in hate speech detection in low-resource NLP research.
- Explored **GAN-based paraphrase generation**, designing differentiable surrogate objectives to address the non-differentiability of discrete text outputs for the generator.

Publications

Peer-reviewed

– Muhammad Umair Haider, **Hammad Rizwan**, Hassan Sajjad, Peizhong Ju, AB Siddique "Neurons Speak in Ranges: Breaking Free from Discrete Neuronal Attribution", 2025, *Initial presentation at Mechanistic Interpretability Workshop at NeurIPS* [PDF](Under Review)

– **Hammad Rizwan**, Domenic Rosati, Ga Wu, Hassan Sajjad "Resolving Lexical Bias in Model Editing" *Proceedings of the 42nd International Conference on Machine Learning (ICML), 2025.* [PDF] (26.9% acceptance rate)

– Ubaid Azam, **Hammad Rizwan**, Asim Karim "Exploring Data Augmentation Strategies for Hate Speech Detection in Roman Urdu." *Proceedings of the Thirteenth Language Resources and Evaluation Conference (LREC), 2022.* [PDF]

– **Hammad Rizwan**, Muhammad Haroon Shakeel, Asim Karim "Hate-Speech and Offensive Language Detection in Roman Urdu." *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2020.* [PDF] (24.5% acceptance rate)

Preprints

- **Hammad Rizwan***, Mahtab Sarvmaili*, Hassan Sajjad, Ga Wu "Instance-Level Difficulty: A Missing Perspective in Machine Unlearning", 2025 [PDF]
- **Hammad Rizwan***, Ubaid Azam*, Faizad Ullah, Asim Karim "Towards Explainable Hate Speech Detection in Roman Urdu" [TBD Link]

Awards and Scholarships

Doctoral Research Scholarship 2023 – 2027

Undergraduate Merit Scholarship (*50% Tuition Coverage*) 2014 – 2017

Invited Talks

Model Interpretability and Model Editing 2025
Deep Learning for NLP – Dalhousie University

Exploration of Neuronal Activations 2024
AI Cluster – Dalhousie University

Computer Vision for Digital Forestry 2022
National Agriculture Robotics Workshop – LUMS

Professional Services

- Co-Organiser: Machine Learning Theory Seminar - Dalhousie University (2024 -)
- Conference Reviewer: NeurIPS 2024 - ; ICML 2025 - ; ICLR 2025 -
- Journal Reviewer: LRE, IPM, NLE